



Epidemiology and patterns of gymnastics-related head & neck trauma injuries: A NEISS database study

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ABSTRACT

Objective: To describe the epidemiology and patterns of gymnastics-related Head & Neck trauma injuries using the NEISS database from 2001 to 2020.

Study design and setting: Cross-sectional analysis of a national database.

Methods: Gymnastics-related ED visits between 2001 and 2020 were queried from the NEISS database. Bivariate chi-squared analyses were used to compare injury demographics, location, type, and disposition. Fracture location was identified using the narrative description of each case and were divided into subtypes for further analysis.

Results: 1455 gymnastics-related head and neck traumatic injuries were identified. The majority were in females (65.8%). The most common presenting age group was pediatric (≤ 18 years) (92.7%), and the largest racial group was Caucasian (51.5%). Of all location subtypes, facial injuries were the most common presenting injury type overall (45.2%). Regarding injury types, lacerations were most common (36.8%), followed by dental injury (30.7%) and fractures (21.2%). The most common location of head and neck fractures was the nose (45.8%), followed by cervical spine (16.7%) and orbit (13.3%). The majority (95.7%) of gymnastics-related head and neck traumatic injuries presenting to the ED were treated and discharged.

Conclusion: This study characterizes gymnastics-related head and neck injuries which is a topic that is understudied. The findings from this study are helpful for gymnasts and those who care for them including providers, coaches and guardians, and this data may help inform future guidelines for treatment and injury prevention.

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1. Introduction

Artistic gymnastics is a physically demanding sport, with both female and male gymnasts competing on a diverse set of events. Women compete on vault, bars, beam and floor [1], while men compete on floor, pommel horse, rings, vault, parallel bars, and the horizontal bar [2]. Successful participation in these events requires a combination of coordination, strength, power, speed, and flexibility [3,4]. Gymnastics is a challenging sport that carries a substantial injury risk. For instance, the window for correct release parameters and successful dismount from the men's high bar has been reported to be ~73–84 milliseconds [5]. Another more well-known example is the phenomenon of losing air space awareness while twisting (colloquially referred to as “the twisties”), experienced by Simone Biles during the Tokyo 2020 Olympic

Games, can render a gymnast unable to land a twisting maneuver properly [6]. As such, it is unsurprising that artistic gymnasts experience high rates of injury.

In comparison with all other NCAA sports, women's gymnastics was found to have the fourth-highest injury rate at practices and competitions [4]. Another study reported injury rates in women's gymnastics were higher than in men's football, and behind only men's wrestling [7]. While gymnastics injuries affect the head and neck infrequently compared to other body parts [4,8,9], trauma to this region may drastically impact quality of life [10].

Recent studies have focused on characterizing head and neck injuries in this athlete population [9,11]. Chorney et al. who studied all maxillofacial injuries among NCAA athletes between 2004 and 2014 reported that gymnastics had one of the lowest overall rate of maxillofacial injuries per 10,000 athlete events during that time [9]. While Bayllis et al., who interviewed 98 gymnasts regarding maxillofacial injuries, reported that the majority of injuries occurred during training and the severity of the injury was strongly associated with the skill level of the gymnast [11]. However, no study to date has comprehensively

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characterized nationwide emergent gymnastics-related head and neck injuries.

Data from the National Electronic Injury Surveillance System (NEISS) [12] has been used to characterize emergent head and neck injuries in other sports [13–16]. This study aims to query the NEISS database to understand the characteristics of head and neck trauma in gymnastics with the goal of providing data that can be used to aid providers, coaches, and guardians who care for gymnasts and may help inform future guidelines for treatment and injury prevention.

2. Methods

An analysis of the Gymnastics-related head and neck trauma was conducted using the NEISS database. Maintained by the Consumer Product Safety Commission (CPSC) since the year 2000, NEISS is comprised of data on consumer product-related injuries collected retrospectively from the emergency departments of 100 hospitals around the United States selected as a probability sample of all 5000+ U.S. hospitals with emergency departments. This database provides pertinent demographics (age, sex, and race), injury information (location, type, and disposition), and a short descriptive narrative of each emergency department (ED) presentation.

Data from the NEISS databases from 2001 to 2020 were combined. Eligibility criteria for this study were gymnastics-related cases of all ages of head and neck traumatic injuries (laceration, fracture, concussion, contusion/abrasion, and dental injury). These cases were identified by using the Product, Body Part and Diagnosis variables within the database. The locations of fracture were identified using the narrative description of each case and were divided into subtypes (nasal, mandibular, midface, cervical spine and skull) for further analysis. National estimates were calculated using the weight variable in the NEISS database.

SPSS v27 (IBM, Armonk, NY) and GraphPad Prism 9 (GraphPad Software, San Diego, CA) software were used for data analysis and data visualization, respectively. Bivariate chi-squared analyses were performed to compare categorical variables. Statistical significance was defined as $p < 0.05$. Because this study utilized a publicly available database maintaining anonymity, it did not require institutional review board approval.

3. Results

Using our criteria, 1455 gymnastics-related head and neck traumatic injuries were identified from NEISS databases (national estimate = 39,042) over a 20-year period (2001–2020). This approximately accounts for 73 injuries/year (national estimate = 1952 injuries/year). The majority of patients presenting with gymnastics-related head and neck trauma were females (65.8%). The average age of presenting patients was 11.74 years (SD = 7.56 years). The pediatric age group (≤ 18 years) comprised an overwhelming majority of patients (92.7%), and age group over 64 years old had no patients. Among those who

Table 1
Demographics of ED patients presenting with gymnastics-related head and neck injuries.

		N (%)
Age group	Pediatric (≤ 18 years)	1349 (92.7%)
	Young adult (19–35 years)	84 (5.8%)
	Middle aged (36–64 years)	22 (1.5%)
	Elderly (> 64 years)	0 (0%)
	Not stated	440 (30.2%)
Race	Caucasian	752 (51.7%)
	African American	149 (10.2%)
	Hispanic	86 (5.9%)
	Asian	24 (1.6%)
	Pacific Islander	3 (0.2%)
	American Indian/Alaska Native	1 (0.1%)
	Not stated	440 (30.2%)

Table 2
Characteristics of gymnastics-related head and neck injuries.

		N (%)
Injury Location	Face	658 (45.2%)
	Head	525 (36.1%)
	Eyeball	17 (1.2%)
	Mouth	187 (12.9%)
	Neck	58 (4%)
	Ear	10 (0.7%)
Injury Type	Laceration	536 (36.8%)
	Concussion	120 (8.2%)
	Fracture	308 (21.2%)
	Contusion/abrasion	45 (3.1%)
	Dental injury	446 (30.7%)
Disposition	Discharged	1393 (95.7%)
	Admitted	56 (3.8%)
	AMA	6 (0.4%)

stated their race, the largest racial group was Caucasian (51.5%) (Table 1).

Facial injuries were the most common presenting injuries overall (45.2%), followed by unspecified head injuries (36.1%), and mouth injuries (12.9%). In terms of injury type, laceration was the most common injury type (36.8%) followed by dental injuries (30.7%). Overall, the vast majority of injuries presenting to the ED were treated and discharged (95.7%), while 3.8% were admitted and 0.4% left against medical advice (AMA) (Table 2).

When studying patients presenting with gymnastics-related head and neck fractures, as shown in Table 3, the most common location of head & neck fractures was found to be the nose (45.8%), followed by cervical spine (16.7%) and orbit (13.3%). The most common injury to the head was concussion (67.8%). Laceration was the most common type of injury to the ear (90.0%), mouth (70.6%), and face (52.2%). Contusion/abrasion was the most common injury to the neck (65.5%) and the only type of injury that the eyeball sustained (Fig. 1).

4. Discussion

Although head and neck injuries occur less commonly in gymnastics than in contact sports [4], these injuries can have substantial consequences. Despite having the potential to cause a major impact on quality of life, gymnastics injuries are generally under-characterized in the literature. This study raises awareness about the prevalence of head and neck injuries among gymnasts.

Our findings showed that injuries to the head represent over one third of head and neck gymnastics-related injuries, with concussions as the most common injury type. TBI and its related syndromes (e.g., chronic traumatic encephalopathy) are becoming a greater part of the public health discourse around sports [17]. Although USA Gymnastics has published a concussion protocol document [18], and there are several published return-to-play guidelines [19], such protocols are not always followed by athletes and/or their coaching staff [19]. Interestingly, given that gymnastics-related injuries are most frequently seen in the pediatric population, organizational and guardian involvement in enforcing concussion protocols warrant a great deal of scrutiny. There are examples anecdotally and in the literature of gymnasts in the

Table 3
Characteristics of gymnastics-related head and neck fractures.

		N (%)
Fracture location	Nasal Bone	55 (45.8%)
	Midface	9 (7.5%)
	Skull	9 (7.5%)
	Orbital	16 (13.3%)
	Mandibular	7 (5.8%)
	Maxillary	4 (3.3%)
	Cervical Spine	20 (16.7%)
	Not stated	440 (30.2%)

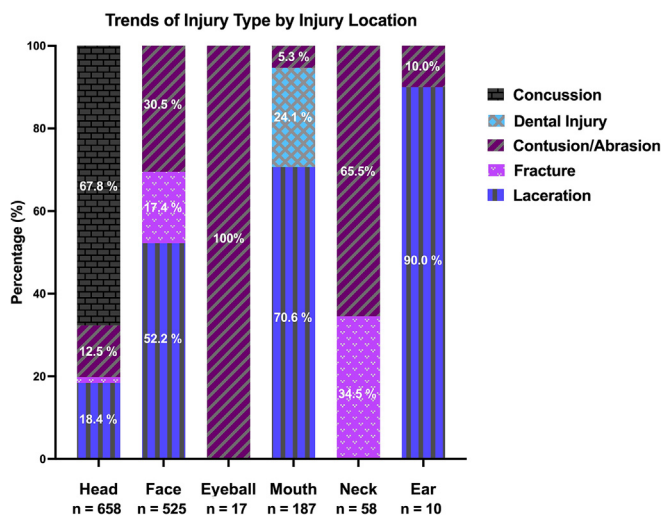


Fig. 1. Gymnastics-related injury type by location in the head and neck.

pediatric population being returned to competition too soon after a traumatic brain injury (TBI), only to result in further episodes of TBI [19]. Due to the fact that concussions represent the most common injury type of gymnastics-related head injuries (demonstrated in our study) and that USA Gymnastics considers the sport “high risk” for concussions [18], it is imperative that guardians, gymnastics organizations, and medical providers be educated and facile at implementing guidelines regarding returning to sport after concussions or mild TBI. Such education may help to mitigate the downstream sequelae of a significant proportion of gymnastics-related head injuries.

It is reassuring that only a small number of patients presenting with head and neck injuries due to gymnastics require admission for further management. However, there remain a high proportion of fracture types, such as cervical spine (16.7%) and orbital (13.3%) that have the potential for catastrophic injury and the need of specialist care [20,21]. Interestingly, it has been shown previously that females who specialized early in their athletic careers—as is common in gymnastics—were at an overall increased risk for injury [22]. It is imperative that proper safety precautions are in place for all skill levels, to minimize severe injury types.

This study is not without limitations. One of these main limitations is the retrospective nature of the NEISS database and its reporting biases. While NEISS compiles reports from a nationally representative probability sample of hospitals [12], it is not exhaustive. Unreported injuries could have different characteristics and frequencies seen in the NEISS database. Furthermore, the database only includes data on emergency department visits, rather than hospital admissions, managements such as surgical interventions or comorbidities of presenting cases. Thus, it might over- or under-estimate particular injury types. It also cannot be used to comment on differences in overall management strategies or outcomes, which would add important and clinically relevant granularity to the data.

5. Conclusion

This study characterizes gymnastics-related head and neck injuries as reported to the NEISS database between 2001 and 2020. These injuries are most commonly seen in the pediatric population. Concussions are the most frequent type of head injuries, and lacerations were the most common overall injury type. Information on gymnastics-related head and neck injuries from this study can be used to improve the care of gymnasts with head and neck injuries, devise strategies for injury prevention, and to promote proper recovery and safe return to sport.

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Contributions

Hisham Abdou: Conception and design of this work, data acquisition, statistical analysis, drafting the manuscript, analysis and interpretation of the data, critical revision of the manuscript for important intellectual content.

Sarah G. Wilkins: Conception and design of this work, drafting the manuscript, analysis and interpretation of the data, critical revision of the manuscript for important intellectual content.

Amar H. Sheth: Conception and design of this work, data acquisition, analysis and interpretation of the data, critical revision of the manuscript for important intellectual content.

Parsa Salehi: Analysis and interpretation of the data, critical revision of the manuscript for important intellectual content.

Yan Ho Lee: Analysis and interpretation of the data, critical revision of the manuscript for important intellectual content.

CRediT authorship contribution statement

Hisham Abdou: Writing – review & editing, Writing – original draft, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Sarah G. Wilkins:** Writing – review & editing, Writing – original draft, Project administration, Methodology, Investigation, Formal analysis, Data curation. **Amar H. Sheth:** Writing – review & editing, Writing – original draft, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Parsa P. Salehi:** Writing – review & editing, Writing – original draft, Project administration, Methodology, Investigation, Conceptualization. **Yan Ho Lee:** Writing – review & editing, Writing – original draft, Supervision, Project administration, Methodology, Investigation, Data curation, Conceptualization.

Declaration of competing interest

The authors have no conflicts of interest to disclose.

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Data access and responsibility: Mr. Abdou, Ms. Wilkins and Mr. Sheth had full access to all the data in the study and take responsibility for the integrity and accuracy of the data analysis.

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